

Use Case “Pay Order by QR”

1. Use case code

UC005

2. Brief Description

This use case describes the interaction between the Customer and the AIMS system when the customer wishes to pay for a placed order using a QR code (VietQR).

3. Actors

3.1. Customer

3.2. VietQR

3.3. AIMS Software

4. Preconditions

1. Customer has successfully placed an order and entered valid delivery information
2. The cart is valid, and stock is sufficient for all selected items.
3. The invoice (product list, VAT, and shipping fees) has been successfully generated
4. The AIMS system is connected to the VietQR sandbox environment.

5. Basic Flow of Events

6. The software displays the payment options and invoice to the Customer, including product list, quantity, total price, VAT, and delivery fee.
7. The Customer chooses a payment method.
8. The software checks the selected method. If [scan QR method = true], the system continues this use case.
9. The software sends a request to VietQR to generate a payment QR code, containing order information (order ID, total amount, merchant code, transaction content).
10. VietQR generates the QR code and returns the encoded image and transaction reference to AIMS software.
11. The software displays the generated QR code on the screen for the Customer to scan.
12. The Customer scans the QR code using a banking app and confirms the payment transaction.
13. The software verifies the payment transaction status through the VietQR callback or query API.

14. The software evaluates the transaction result
15. The software updates the order status to *Pending Processing*, records the transaction, and sends the payment confirmation email to the Customer.
16. The software displays a message confirming successful payment and summarizes order and transaction information.

6. Alternative flows

Table N-Alternative flows of events for UC Pay order by QR

No	Location	Condition	Action	Resume location
1	Step 4	VietQR API fails to generate QR code.	▪ Display “Unable to generate QR code.” ▪ Allow retry or switch to another payment method.	Step 2
2	Step 7	Customer cancels or closes the payment app before confirming.	▪ Display “Payment not completed.” ▪ Offer retry option.	Step 2
3	Step 8	Payment callback not received (timeout or network error).	▪ Retry verification after a fixed interval. ▪ If still not confirmed, mark order as “Awaiting Payment.”	Use case ends
4	Step 9	Payment verification fails (invalid signature or mismatch).	▪ Reject transaction and notify customer. ▪ Mark order as “Payment Failed.”	Use case ends
5	Step 10	Inventory update or database error	▪ Log the error and set order as “Pending Manual Review.”	Use case ends

7. Input data

Table A-Input data of UC005

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Order ID	Unique identifier of the customer's order	Yes	Alphanumeric	ORD-2025-1008-0012
2.	Total amount	Total payable (subtotal + VAT + shipping)	Yes	≥ 0	215,000 VND
3.	Merchant code	Store identifier for VietQR	Yes	String	AIMS-HUST-001
4.	Transaction content	Description for payment record	Yes	≤ 255 chars	Payment for Order ORD-2025-1008-00
5.	Customer info	Customer name, phone, email, address	Yes	Valid format	Nguyen Van A, 0901234567, Hanoi
6.	QR content	Encoded payment string returned from VietQR	Yes	Must follow VietQR format	https://api.vietqr.vn/qr/ab23xyz

Output data**Table B-Output data**

No	Data fields	Description	Display format	Example
1.	QR code image	Image for customer to scan	PNG/Base64	
2.	Payment confirmation message	Text showing payment success	Text	"Payment completed successfully via VietQR."
3.	Transaction details		Text / Table	TXN-998877 – 215,000 VND – 2025-10-08 18:36
4.	Order status	Update order state	Text	"Pending Processing"
5.	Confirmation email	Invoice and transaction summary sent to customer	Email (HTML/PDF)	

9. Postconditions

- Payment transaction is successfully confirmed through VietQR.

- Order status is updated to **Pending Processing**.
- Product inventory is decreased according to the order quantity.
- The customer receives an email with the invoice and payment confirmation.
- AIMS stores the transaction record for reconciliation and auditing purposes.

Use Case “Pay Order by Credit Card”

1. Use case code

UC006

2. Brief Description

This use case describes how the Customer completes an online payment through PayPal (credit card) in the AIMS system.

3. Actors

3.1. Customer

3.2. VietQR

3.3. AIMS Software

4. Preconditions

1. The customer has completed checkout and is viewing a valid invoice.
2. Stock availability and total payable amount are confirmed.
3. The AIMS system is configured with valid PayPal Client ID/Secret and connected to the PayPal Sandbox API.
4. Network connectivity between AIMS and PayPal is stable.

5. Basic Flow of Events

1. The software displays payment options and invoice.
2. The Customer selects Credit Card (PayPal) as the payment method.
3. The software validates order data and reserves stock.
4. The software obtains a PayPal OAuth token and creates a PayPal order.
5. PayPal returns an approval URL.
6. The software redirects the Customer to PayPal.
7. The Customer logs in or enters card details and approves payment.
8. PayPal redirects the Customer back to AIMS with transaction info.
9. The software verifies and captures the PayPal order.
10. If capture successful, AIMS stores transaction data, updates stock and order status, and sends a confirmation email.
11. If capture failed, AIMS cancels the draft order and releases reserved stock.
12. The software displays payment result to the customer.

6. Alternative flows

Table N-Alternative flows of events for UC Pay order by QR

No	Location	Condition	Action	Resume location
1	Step 7	Customer cancels payment or closes PayPal before approval.	▪ Display “Unable to generate QR code.” ▪ Allow retry or switch to another payment method.	Step 2
2	Step 4	PayPal API or token request fails.	▪ Display “Payment not completed.” ▪ Offer retry option.	Step 2
3	Step 9	Capture not completed or declined.	▪ Display “Payment unsuccessful.” ▪ Roll back draft order and release stock.	Step 2
4	Step 10	Error while updating order or stock.	▪ Log error, mark order “Pending Manual Review.”	Use case ends

7. Input data

Table A-Input data of UC005

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Order ID	Unique order identifier	Yes	Alphanumeric	ORD-2025-1008-0012
2.	Total amount	Sum of items + VAT + shipping	Yes	≥ 0	215,000 VND
3.	Customer info	Name, phone, email, address	Yes	Valid format	Nguyen Van A, 0901234567, Hanoi
4.	PayPal OAuth Token	Authentication token	Yes	Valid JWT format	A21AAKQ...token
5.	Approval URL	Redirect link returned by PayPal	Yes	HTTPS URL	https://www.sandbox.paypal.com/...

6.	Capture ID	Identifier of successful capture	Yes	Alphanumeric	CAP-9AB23952RC1234567
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8. Output data

Table B-Output data of UC005

No	Data fields	Description	Display format	Example
1.	Approval link	Redirect URL for PayPal checkout	Hyperlink	
2.	Transaction status	Text showing payment success	Text	“Payment completed successfully via Paypal.”
3.	Transaction info	Capture ID, amount, payer, time	Table	CAP-9AB23952RC1234567 – 325,000 VND – 2025-10-08 18:42
4.	Order status	Update order state	Text	“Pending Processing”
5.	Confirmation email	Invoice and transaction summary sent to customer	Email (HTML/PDF)	

9. Postconditions

- The PayPal transaction is successfully captured and recorded.
- Order status is updated to Pending Processing.
- Inventory quantities are adjusted.
- Confirmation email is sent to the customer.
- In case of failure or cancellation, the system releases stock and keeps an audit log.